

TEK4090

Introduction to Modern Control

Lecture 5: Estimation

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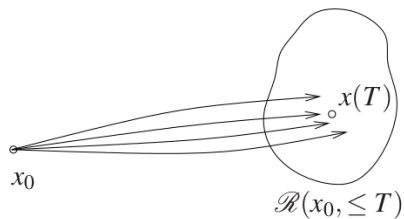
Recap from Last Time

- ▶ Pole placement
- ▶ Optimal control
- ▶ Linear Quadratic Regulator
- ▶ MPC and Building Control

Reading Assignment

- ▶ **Åstrom & Murray:** Ch. 7
- ▶ **Crassidis & Junkins:**
 - ▶ Kalman filters: Ch. 3.1–3.3
 - ▶ Spacecraft Dynamics: Ch. 6.1, Appendix A.7-A.8
 - ▶ Probability: Appendix C

Reachability



The **reachable set** $\mathcal{R}(x_0, T)$ is the set of all points x_f such that there exists an input $u(t)$ that steers the system from $x(0) = x_0$ to $x(T) = x_f$.

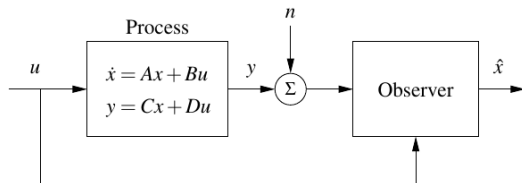
A system is **reachable** if for any $x_0, x_f \in \mathbb{R}^n$, there exists $T > 0$ and $u : [0, T] \mapsto \mathbb{R}$ such that $x(0) = x_0$ and $x(T) = x_f$.

Observability

Feedback requires a measurement of the system.

How do we use measurements of a system in the control loop?

- ▶ Consider the LTI system (A, B, C, D)
- ▶ Suppose y is corrupted by noise n
- ▶ We can construct an **observer** to produce the estimate $\hat{x}$



Observability

Challenges/Questions:

1. How 'rich' does our measurement need to be in order to be useful?
2. How do you design an observer?
3. How does the interconnection between the observer and the controller work?

Observability

Answer to **Challenge 1**:

Definition:

A system $\Sigma := (A, B, C, D)$ is said to be **observable** if, for any $T > 0$, it is possible to reconstruct the state $x(T)$ from measurements of $y(t)$ and $u(t)$ on $[0, T]$.

So, when is a system observable?

Observability

It's useful just to consider the autonomous system

$$\dot{x} = Ax$$

$$y = Cx.$$

Immediately note: if C is invertible, then we can recover $x(t)$ from $x(t) = C^{-1}y(t)$.

In general, C is rank-deficient and can have fewer rows than x .

Observability

Observability

Observability

Theorem:

The system $\Sigma := (A, B, C, D)$ is observable if and only if the observability gramian W_o has full rank.

Observable Canonical Form

$$\dot{z} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} z + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0 \quad \cdots \quad 0] z + Du$$

Observable Canonical Form

$$W_o = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -a_1 & 1 & 0 & \cdots & 0 \\ -a_1^2 - a_2 & -a_1 & 1 & & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ * & * & & \cdots & 1 \end{bmatrix}$$

$$W_o^{-1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a_1 & 1 & 0 & \cdots & 0 \\ a_2 & a_1 & 1 & & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \cdots & 1 \end{bmatrix}$$

Observers & State Estimation

Answer to Challenge 2:

- ▶ Design a dynamical system for $\dot{\hat{x}}$
- ▶ Make $\hat{x}(t) \rightarrow x(t)$ as $t \rightarrow \infty$

Observers & State Estimation

Make a copy of the system and propagate the estimate with the right input

Original System

$$\dot{x} = Ax + Bu$$

Copy of System

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

Observers & State Estimation: Error Dynamics

Observers & State Estimation: Error Dynamics

Duality

Controller:

$$A - BK$$

$$A \leftrightarrow A^T$$

$$B \leftrightarrow C^T$$

Observer:

$$A - LC$$

$$L \leftrightarrow L^T$$

$$W_r \leftrightarrow W_o^T$$

(A, B) reachable iff (A^T, B^T) observable; (A, C) observable iff (A^T, C^T) reachable

Observer Design by Eigenvalue Assignment

$$\frac{dx}{dt} = Ax + Bu$$

$$y = Cx$$

$$\frac{d\hat{x}}{dt} = A\hat{x} + Bu + L(y - C\hat{x})$$

$$\lambda(s) = s^n + a_1s^{n-1} + \dots + a_n$$

$$L = W_o^{-1} \tilde{W}_o \begin{bmatrix} p_1 - a_1 \\ \dots \\ p_n - a_n \end{bmatrix}$$

$$\tilde{W}_o = \begin{bmatrix} 1 & 0 & \dots & 0 \\ a_1 & 1 & \ddots & 0 \\ a_2 & a_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-2} & a_{n-3} & \dots & 0 \\ a_{n-1} & a_{n-2} & \dots & 1 \end{bmatrix}^{-1}$$

$$p(s) = s^n + p_1s^{n-1} + \dots + p_n$$

Challenge 3:

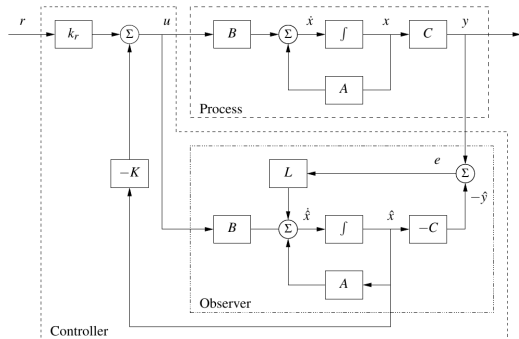
How does the controller work with estimated states?

Control with Estimated State

Control with Estimated State

Separation Principle

The controller and observer gains can be designed **independently** of each other



$$\frac{d}{dt} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} + \begin{bmatrix} Bk_r \\ 0 \end{bmatrix} r$$

$$\lambda(s) = \det(sI - A + BK) \det(sI - A + LC)$$

Kalman Decomposition

Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & 0 & 0 \\ 0 & A_{r\bar{o}} & 0 & 0 \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & 0 & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$

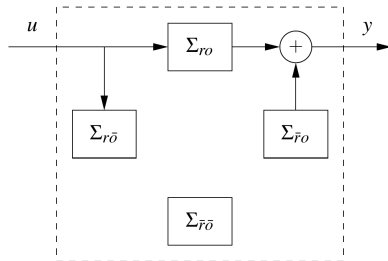
Non-Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & * & 0 \\ * & A_{r\bar{o}} & * & * \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & * & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$

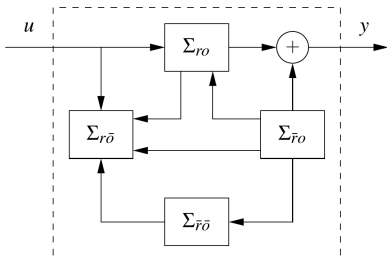
Kalman Decomposition

Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & 0 & 0 \\ 0 & A_{r\bar{o}} & 0 & 0 \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & 0 & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$



Kalman Decomposition



Non-Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & * & 0 \\ * & A_{r\bar{o}} & * & * \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & * & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$

Back to Challenge 2:

We saw optimal gains for control, what about observers?

How to choose the observer gain optimally?

In general, measurements are **noisy**.

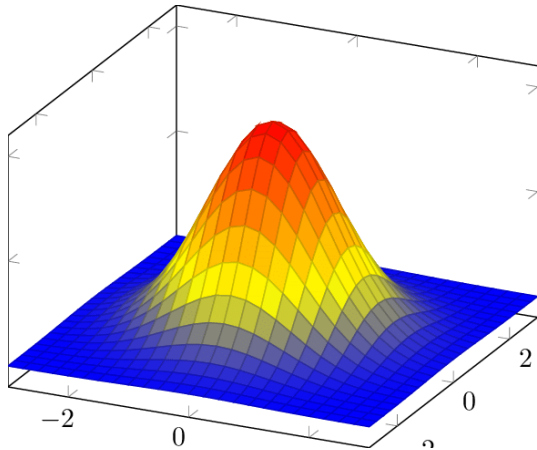
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$

$$y_k = Cx_k + Du_k + w_k$$

$$E[v_k] = 0, \quad E[w_k] = 0$$

$$E[v_k v_{k+1}^T] = \begin{cases} 0 & k \neq j \\ Q_k & k = j \end{cases}$$

$$E[w_k w_{k+1}^T] = \begin{cases} 0 & k \neq j \\ R_k & k = j \end{cases}$$

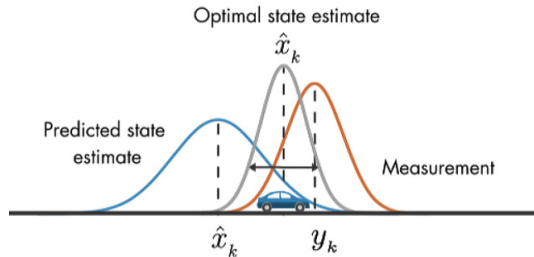


How to choose the observer gain?

With noisy measurements, the estimate is now a **random variable**.

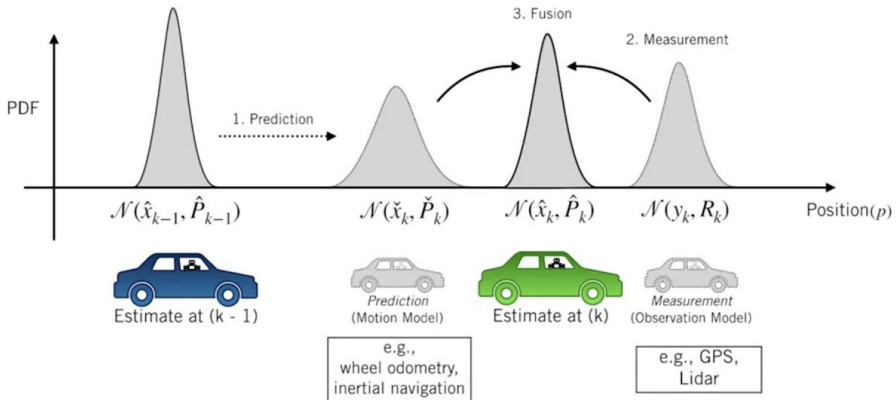
If L stabilizes the observer dynamics, then

$$E[e_k] = E[\hat{x}_k - x_k] \rightarrow 0.$$



Instead, we can optimize the 'width' (or, covariance) of the estimate

Optimal Observers



Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter

Model:
$$\begin{aligned}x_{k+1} &= Ax_k + Bu_k + Fv_k \\y_k &= Cx_k + Du_k + w_k\end{aligned}$$

Initialize: $\hat{x}_0 = x_0, P_0 = P$

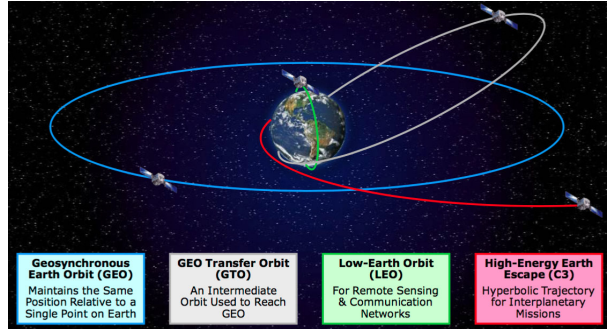
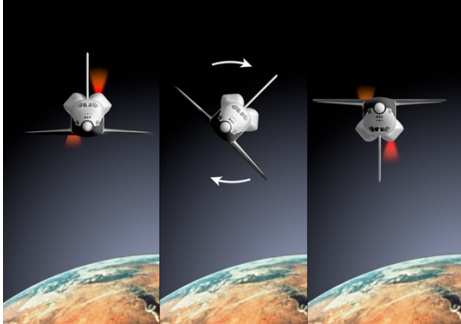
Gain:
$$K_k = P_k^- C^T (CP_k^- C^T + R_k)^{-1}$$

Update:
$$\begin{aligned}\hat{x}_k^+ &= \hat{x}_k^- + K_k (\tilde{y}_k - C\hat{x}_k^-) \\P_k^+ &= (I - K_k C) P_k^-\end{aligned}$$

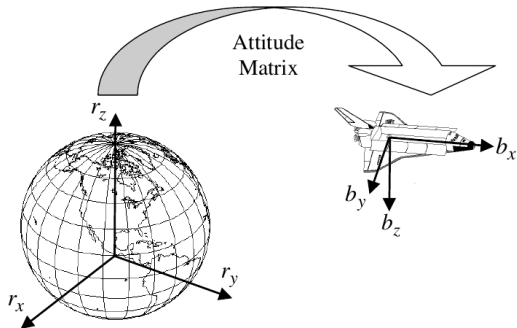
Propagate:
$$\begin{aligned}\hat{x}_{k+1}^- &= A\hat{x}_k^+ + Bu_k \\P_{k+1}^- &= AP_k^+ A^T + BQ_k B^T\end{aligned}$$

Spacecraft GNC

Spacecraft GNC



Attitude



Reference and body frames:

$$b = b_x \hat{b}_1 + b_y \hat{b}_2 + b_z \hat{b}_3$$

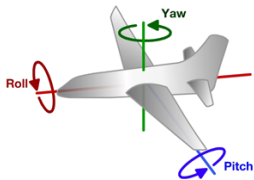
$$r = r_x \hat{r}_1 + r_y \hat{r}_2 + r_z \hat{r}_3$$

Transformation between the two:

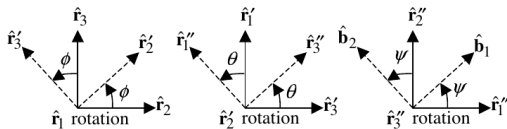
$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = A \begin{bmatrix} \hat{r}_1 \\ \hat{r}_2 \\ \hat{r}_3 \end{bmatrix} \quad (1)$$

Euler Angles

One can represent a rotation via the **Euler angles**: roll ϕ , pitch θ , and yaw ψ .

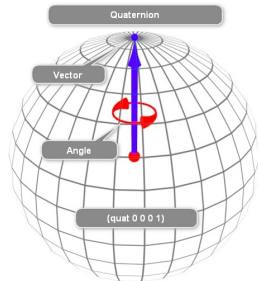


$$r' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} r \quad r'' = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} r' \quad b = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} r''$$



Quaternions

The Euler angles have a singularity when the attitude is pointed at a pole, so spacecraft attitude is represented as a **quaternion** instead.



$$q = \begin{bmatrix} \rho \\ q_4 \end{bmatrix} \in \mathbb{R}^4$$

$$q^T q = 1$$

$$A(q) = \Xi^T(q)\Psi(q)$$

$$\Xi(q) = \begin{bmatrix} q_4 I_{3 \times 3} + [\rho \times] \\ -\rho^T \end{bmatrix}$$

$$\Psi(q) = \begin{bmatrix} q_4 I_{3 \times 3} - [\rho \times] \\ -\rho^T \end{bmatrix}$$

$$[\rho \times] = \begin{bmatrix} 0 & -\rho_3 & \rho_2 \\ \rho_3 & 0 & -\rho_1 \\ -\rho_2 & \rho_1 & 0 \end{bmatrix}$$

Rigid Body Dynamics

Dynamics: Euler Angles

- ▶ H : Angular momentum
- ▶ J : Moment of Inertia
- ▶ L : Applied torque
- ▶ ω : Angular velocity

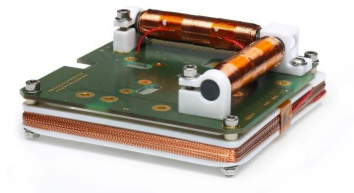
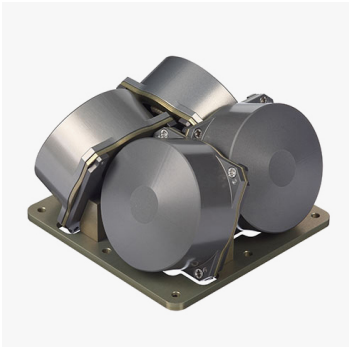
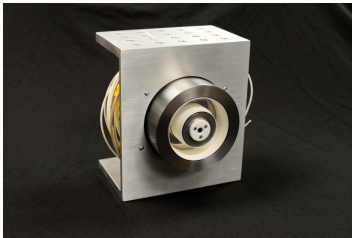
$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega \times]J\omega + L\end{aligned}$$

Kinematics: Quaternions

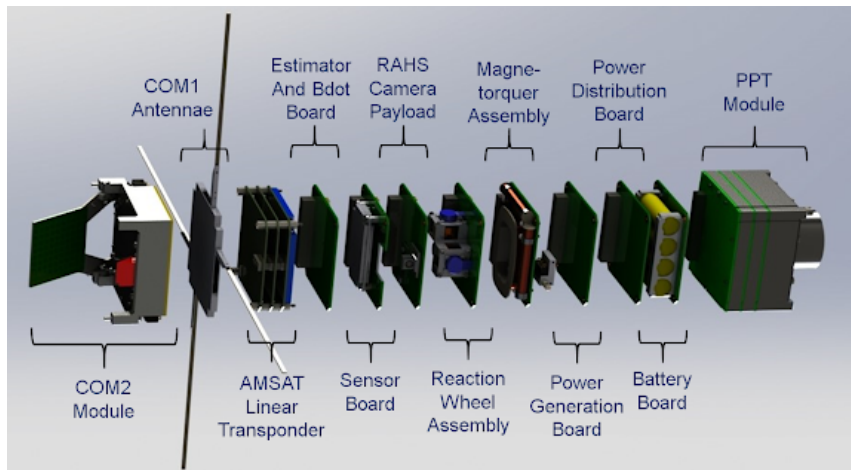
$$\begin{aligned}\dot{q} &= \frac{1}{2}\Omega(\omega)q \\ \Omega(\omega) &= \begin{bmatrix} -[\omega \times] & \omega \\ -\omega^T & 0 \end{bmatrix}\end{aligned}$$

The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

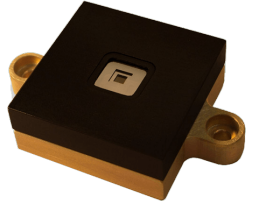
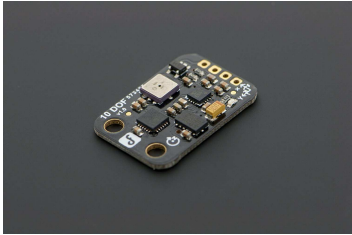
Spacecraft Actuators



Plasma Thruster

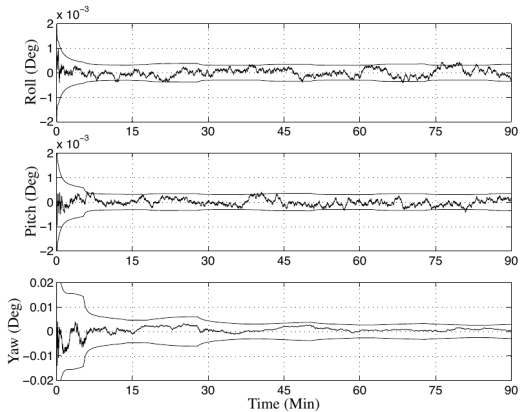


Measurement



Quaternion Extended Kalman Filter

Initialize	$\hat{\mathbf{q}}(t_0) = \hat{\mathbf{q}}_0, \quad \hat{\boldsymbol{\beta}}(t_0) = \hat{\boldsymbol{\beta}}_0$ $P(t_0) = P_0$
Gain	$K_k = P_k^- H_k^T (\hat{\mathbf{s}}_k^-) [H_k(\hat{\mathbf{s}}_k^-) P_k^- H_k^T (\hat{\mathbf{s}}_k^-) + R]^{-1}$ $H_k(\hat{\mathbf{s}}_k^-) = \begin{bmatrix} A(\hat{\mathbf{q}}^-) \mathbf{r}_1 \times & 0_{3 \times 3} \\ \vdots & \vdots \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_n \times & 0_{3 \times 3} \end{bmatrix} \Big _{l_k}$
Update	$P_k^+ = [I - K_k H_k(\hat{\mathbf{s}}_k^-)] P_k^-$ $\Delta \hat{\mathbf{s}}_k^+ = K_k [\bar{\mathbf{y}}_k - \mathbf{h}_k(\hat{\mathbf{s}}_k^-)]$ $\Delta \hat{\mathbf{s}}_k^+ \equiv [\delta \hat{\boldsymbol{\alpha}}_k^{+T} \quad \Delta \hat{\boldsymbol{\beta}}_k^{+T}]^T$ $\mathbf{h}_k(\hat{\mathbf{s}}_k^-) = \begin{bmatrix} A(\hat{\mathbf{q}}^-) \mathbf{r}_1 \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_2 \\ \vdots \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_n \end{bmatrix} \Big _{l_k}$ $\hat{\mathbf{q}}_k^+ = \hat{\mathbf{q}}_k^- + \frac{1}{2} \Xi(\hat{\mathbf{q}}_k^-) \delta \hat{\boldsymbol{\alpha}}_k^+, \quad \text{re-normalize quaternion}$ $\hat{\boldsymbol{\beta}}_k^+ = \hat{\boldsymbol{\beta}}_k^- + \Delta \hat{\boldsymbol{\beta}}_k^+$
Propagation	$\dot{\boldsymbol{\omega}}(t) = \hat{\boldsymbol{\omega}}(t) - \hat{\boldsymbol{\beta}}(t)$ $\dot{\hat{\mathbf{q}}}(t) = \frac{1}{2} \Xi(\hat{\mathbf{q}}(t)) \hat{\boldsymbol{\omega}}(t)$ $\dot{P}(t) = F(t) P(t) + P(t) F^T(t) + G(t) Q(t) G^T(t)$ $F(t) = \begin{bmatrix} -[\hat{\boldsymbol{\omega}}(t) \times] & -I_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}, \quad G(t) = \begin{bmatrix} -I_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix}$



Matlab - simulating ODEs

```
1 clear all; close all; clc;
2
3 % simulation timespan
4 dt=0.1;
5 t_space = [0:dt:10];
6
7 % initial condition
8 x_0 = [10;10];
9
10 % sinusoidal input with zero-order-hold
11 u = cos(t_space);
12 x_array = zeros(2, length(t_space));
13
14 x_array(:,1) = x_0;
15
16 for ii = 1:length(t_space)-1
17     t_int = [t_space(ii), t_space(ii+1)];
18
19     %simulate ODE with input
20     [t_out,y_out] = ode45(@fn_rhs,t_int,x_array(:,ii),[],u(ii));
21     x_array(:,ii+1) = y_out(end,:);
22
23 end
24
25
26 plot(t_space, x_array(1,:)), hold on
27 plot(t_space, x_array(2,:))
```

```
1 function [x_dot] = fn_rhs(t, x, u)
2 %fn_rhs: a 'right-hand-side function' of the ode x_dot = f(t,x,u)
3
4 x_dot = exp(-t)*[cos(x(1)); sin(x(2))] + exp(-0.1*t)*[0;1]*u;
5
6 end
```